engine. By doing this, we would not only gain hands-on knowledge and skills, we could also tweak the engine to suit our future needs. L-17 did not appeal to me for another reason: it was a pressure-fed engine, which was not ideal to work upon for futuristic missions. I wanted a turbo pump-fed engine with higher thrust level.

The next candidate was a liquid engine called M-40, which had a thrust of 40 tonnes. It used nitrogen tetroxide (N2O4) and unsymmetrical dimethyl hydrogen (UDMH) as fuel. Morin said they were planning a 55-tonne thrust rocket and wanted to upgrade M-40 for that. Grover, a constant companion in our dealings with SEP, realised the potential for a France-India tie-up on this. When he proposed this to SEP chairman Souffle, he dismissed it saying Indians wouldn't have the expertise to work with them. That was a notion that the ISRO team of scientists would soon demolish to show how Indian engineers can be as good, if not better, than anyone if given an opportunity. Grover went to great lengths to convince Souffle that though ISRO had not developed big rocket engines, its scientists were much more than a bunch of greenhorns. There were young scientists who had passed out of IITs, our premier engineering institutes, Grover reasoned. He asked Souffle if he considered me, a graduate of Princeton University, as any less capable than their scientists. Souffle appeared convinced.

I shuttled between Trivandrum and Paris for the next three years, trying to firm up the collaboration. Parallel to this, Sarabhai had been trying for a German joint venture, which did not take off. And Sarabhai did not live to see the results of our French connection: the Vikas engine.

Even as I was working on the French option, Sarabhai was talking to some German scientists, especially a professor called Prof. Armin Dadieu. The Germans were masters of metallurgy and fabrication. Germany's has been an unfortunate degeneration in the field of space and missile technology after its defeat in World War II. As early as the early 1940s, Germany under Hitler had developed a liquid rocket called V-2. Hitler wanted to use V-2 with a warhead. In other words, as a missile. But there was a hitch: V2 did not have a guidance system to hit targets. The Germans still gave England the scare, randomly firing V-2 with warhead across the